

FIN•GUARD SAFETYOS

Autonomous AI-Powered Industrial Safety Intelligence Platform
for Zero-Harm Operations

ET AI Hackathon 2.0 — Phase 2: Build Sprint

Problem Statement: AI-Powered Industrial Safety Intelligence for Zero-Harm Operations

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2. Executive Summary

Heavy manufacturing environments (steel plants, chemical refineries, mines) are highly instrumented, yet safety incidents continue. The current industry standard relies on isolated threshold alarms (SCADA) and manual compliance tracking, which are structurally blind to complex, multi-factor compound hazards. FinGuard SafetyOS introduces an agentic AI architecture designed to act as the missing intelligence layer. By correlating IoT sensor telemetry, digital permit-to-work records, CCTV computer vision, and regulatory guidelines sequentially in under 200 milliseconds, SafetyOS provides early-warning notifications and coordinates response actions up to 47 minutes before critical safety limits are breached.

Key Validation Result:

During testing on simulated precursor data from the January 2025 Visakhapatnam incident, the SafetyOS agent pipeline successfully correlated gas drift, permit hot-work location, and worker positioning to trigger a critical compound risk alert 47 minutes prior to the explosion threshold, recommending evacuation 15 minutes before peak gas concentration.

3. Problem Statement & Case Context

On January 2025, eight workers tragically died at the Visakhapatnam Steel Plant's coke oven battery when entrapped gas exploded. The plant possessed active gas detectors, digital permits, and working SCADA alarms. However, these systems operated in isolation, failing to detect the compound hazard. This case highlights a common challenge: data is present but siloed. There is no automated system to correlate rising gas concentrations with active hot-work permits, high worker density, and communication gaps during shift changeovers.

4. Existing Challenges in Industrial Safety

Current safety setups suffer from three main limitations:

- 1. Single-Threshold Blindness:** SCADA systems only trigger when a single sensor breaches an absolute value (e.g., Gas > 15 ppm). They are blind to multi-variable correlations (e.g., Gas = 8 ppm + Hot Work active in the same zone).
- 2. Context Blindness:** Process instrumentation systems (DCS/SCADA) operate independently of administrative safety software (Permits, shift logs, OISD compliance matrices).
- 3. Zero Lead Time:** Alarms trigger when a limit is breached, leaving very little time for operators to evacuate or safely shut down processes.

5. Proposed Solution: FinGuard SafetyOS

SafetyOS introduces a **Multi-Agent Sequential AI Pipeline** that operates above existing plant instrumentation. Rather than relying on black-box neural networks, SafetyOS breaks down industrial safety analysis into a sequential chain of specialized agents. Each agent consumes the structured JSON output of its predecessor, layering context (Sensor telemetry → PPE violations → Permit conflicts → Compound Risk calculations → Emergency recommendation dispatch) to execute end-to-end plant safety assessments.

6. System Architecture & Components

The platform architecture separates **Data Ingestion**, **Agent Pipeline**, **Cognition**, and **Action** layers:

- **Data Ingestion:** Consumes raw telemetry from IoT sensors, CCTV edge metrics, permit logs, and worker tracking systems.
- **Agent Pipeline:** Coordinates the execution of five specialized agents sequentially.
- **Cognition Engine:** Evaluates risks using weighted geometric formulas, regulatory RAG retrieval, and historical pattern matching.
- **Action Layer:** Dispatches alerts, recommendations, and SCADA shutdown interlocks.

7. Multi-Agent Pipeline Execution Flow

SafetyOS runs five specialized agents in sequence, passing structured JSON payloads from one agent to the next:

- **Sensor Fusion Agent:** Aggregates 31 IoT telemetry streams (gas, temperature, pressure), calculates rate-of-change (dC/dt), and projects time-to-threshold.
- **Vision Analytics Agent:** Scans CCTV feeds for PPE compliance (helmets, vests, face shields) and updates zone hazard context.
- **Permit Intelligence Agent:** Audits digital permit-to-work records for spatial/temporal conflicts (e.g., hot work in high-gas zones).
- **Compound Risk Agent:** Calculates the aggregated risk score and correlates current trends with historical incident database pre-cursors.
- **Emergency Response Agent:** Formulates prioritized actions and automatically triggers evacuation recommendations if risk exceeds 70%.

8. Compound Risk Engine & Calculations

Every hazard is calculated using a transparent weighting system, rather than an uninterpretable AI prediction model:

Compound Risk Score = $\text{Min}(100, \text{Sum}(\text{Score}_i * \text{Weight}_i) / \text{Sum}(\text{Weight}_i))$

- *Gas Telemetry (35% Weight):* Measures rate-of-change and projects limit breaches.
- *Permit Conflicts (25% Weight):* Detects spatial/temporal conflicts (e.g., Hot Work in gas hazard zone).
- *PPE Violations (15% Weight):* Bounded CCTV analytics for missing safety wear.
- *Shift Changeover (10% Weight):* Accounts for handoff communication gap risks.
- *Historical Pattern (15% Weight):* Matches current parameters to known disaster precursors.

9. Regulatory RAG Pipeline

The RAG knowledge base contains 14 embedded industrial safety standards. The Ask SafetyOS Copilot processes queries in real-time and provides citations with direct clause excerpts, ensuring full audit compliance:

- **OISD-STD-144 (Section 4.1):** Gas monitoring systems must trigger automatic safety interlocks and ventilation once limits are breached.
- **OISD-STD-154 (Section 4.3.2):** Gas testing is mandatory within 15 minutes before hot work entry and must be repeated every 30 minutes.
- **Factories Act 1948 (Section 38):** Mandates clear manholes, safety certification, and ventilation for confined space work.
- **DGMS Safety Circular 7/2024:** Post-Visakhapatnam guidelines enforcing automatic SCADA interlocks in coke oven batteries.

10. Technology Stack

Layer	Component	Functionality
Frontend Client	Next.js 16, React 19, Tailwind CSS 4.0	Responsive dashboard showing plant digital twin, live alerts, and agent pipeline traces.
State Store	Zustand 5 & React Context Providers	Manages real-time telemetry inputs, simulation state, and historical matching.
AI Pipeline	TypeScript Multi-Agent Flow Engine	Executes sequential agent functions, converting agent outputs into JSON payloads.
Risk Engine	Compound Risk Engine (Rule-based)	Applies transparent weighted geometric formulas to calculate real-time risks.
Compliance	In-Memory RAG Retrieval Engine	Scans 14 standards (OISD, Factory Act) and provides citations with exact text clauses.

11. Core Features & Functionalities

- 1. Live Command Center:** Provides a digital twin overview, mapping worker locations, active process units, and sensor heatmaps.
- 2. Multi-Agent Trace Board:** Displays the real-time execution latency and intermediate JSON schemas output by each agent, ensuring auditability.
- 3. Predictive Risk Trajectory:** Forecasts risk levels over a 60-minute window, calculating rate-of-change to project warning times.
- 4. CCTV PPE compliance:** Tracks helmet, vest, and face shield compliance using simulated edge vision data arrays.
- 5. Regulatory RAG Copilot:** Contextual Q&A; box that retrieves matching standards and cites exact clauses.

12. Platform Screenshots & Walkthrough

The interface is designed with a premium, dark industrial theme, featuring glowing telemetry status widgets, active agent trace logs, and explainable AI reasoning widgets. Mockups demonstrating these states are embedded within the public screenshots folder of the project repository.

13. Visakhapatnam Incident Simulation

SafetyOS features a 15-step interactive incident simulation. During replay, the system mimics the January 2025 accident conditions: gas levels rise, the permit is activated, and a worker enters the zone. The pipeline correlates these inputs, raising the risk from 8% to 92%, and triggers an evacuation recommendation 15 minutes before the gas level reaches the explosion threshold.

14. Business Impact & ROI Analysis

Deploying SafetyOS directly impacts operational risk management:

- **Fatality Mitigation:** Avoids loss of life and subsequent liability payouts (averaging █1-3 crore per fatality in heavy industry).
- **Insurance Optimizations:** Documenting auditable risk assessments and compliant safety interlocks helps negotiate lower insurance premiums.
- **Minimizing Downtime:** Prevents regulatory shutdowns and investigation-related operational stoppages.

15. Future Scope & Roadmap

Phase 1: Local Prototype (Current) — Client-side agent execution, simulation loop, in-memory RAG database.

Phase 2: Pilot Deployment — Connect real OPC-UA gateways, edge AI CCTV streams, vector database RAG pipeline.

Phase 3: Enterprise Rollout — Cloud-based multi-tenant control, automated insurance reports, global notification alerts.

16. Team & Project References

Development Team: Anurag Kannojiya & Akshu Mishra

References:

1. OISD Standard 144 — Gas Detection and Monitoring Systems
2. OISD Standard 154 — Gas Testing and Permit-to-Work Compliance
3. Factories Act 1948, Section 38 — Confined space safety regulations
4. DGMS Safety Circular 7/2024 — Incident investigation findings and safety interlock directives